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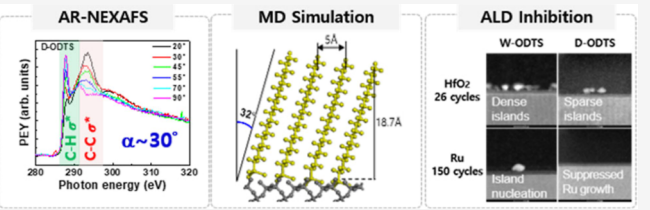
Structure-Inhibition Relationship of Vapor-Deposited Alkylsilane Monolayers in Area-Selective Atomic Layer Deposition

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ABSTRACT: Area-selective atomic layer deposition (AS-ALD) relies critically on molecular-scale inhibition layers to suppress precursor adsorption in nongrowth regions. Although octadecyltrichlorosilane (ODTS) self-assembled monolayers (SAMs) are widely used, their structural parameters—particularly molecular orientation and grafted molecular areal density—have rarely been systematically correlated with inhibition performance. Moreover, conventional solution-phase deposition introduces contamination risks and extended processing times, limiting practical adoption. Here, we demonstrate a rapid vapor-phase deposition route that forms ODTS blocking layers with improved molecular alignment on SiO_2 within 3 min. Angle-dependent near-edge X-ray absorption fine structure (NEXAFS) spectroscopy reveals a reduced molecular tilt angle of ∼29.8 ± 0.3° for vapor-deposited ODTS compared to ∼38.7 ± 0.6° for solution-deposited films, indicating a more upright average molecular orientation and lower orientational disorder. Molecular dynamics simulations establish an inverse correlation between tilt angle and grafted molecular areal density, yielding an estimated density of ∼2 molecules/nm^2 for vapor-deposited layers. When applied as ALD inhibitors, the vapor-deposited ODTS layers exhibit improved blocking behavior for HfO_2 and Ru ALD compared with the solution-deposited ODTS layers prepared in this study. For HfO_2 ALD, image-based analysis at the same projected area fraction shows a ∼35% decrease in nucleation density on D-ODTS, while Ru nucleation is strongly suppressed on D-ODTS, with only isolated nuclei observed within the examined field of view. These results indicate that average molecular orientation and effective grafted molecular areal density are correlated with precursor-accessible nucleation pathways and ALD inhibition efficiency.



1. INTRODUCTION

As semiconductor devices continue to shrink and adopt increasingly complex three-dimensional (3D) architectures, conventional top-down fabrication techniques—comprising iterative deposition, photolithography, and etching—face critical scaling limitations in pattern resolution, misalignment, and the patterning of 3D structures. Area-selective atomic layer deposition (AS-ALD) has emerged as a promising bottom-up alternative, enabling spatially controlled thin-film growth without photolithography or etching steps by exploiting differences in local surface reactivity.[1−3]

The performance of AS-ALD is quantified by selectivity, a metric originally formulated by Gladfelter that measures the disparity in material deposition between growth areas (GA) and nongrowth areas (NGA)[4]

$$\text{selectivity(s)} = \frac{\theta_{GA} - \theta_{NGA}}{\theta_{GA} + \theta_{NGA}}$$

where θ_{GA} and θ_{NGA} represent the amounts of material deposited in the GA and NGA, respectively. AS-ALD integrates surface modification with ALD's inherently self-limiting surface reactions, achieving selectivity by chemically deactivating specific surface sites prior to ALD cycles.[1,4−7]

Beyond AS-ALD, self-assembled monolayers (SAMs) are widely used as molecularly thin interfacial layers for surface functionalization and interfacial engineering in thin-film electronic devices. A recent review highlighted that molecular orientation, packing density, conformational order, and defect density critically determine SAM-mediated interface properties.[8] This broader context motivates a more quantitative understanding of SAM structure−function relationships in AS-ALD.

A widely adopted strategy for site deactivation employs SAM inhibitors.[9−17] In addition to SAM-based approaches, small-molecule inhibitors (SMIs) have also been explored for selective surface passivation, offering advantages such as rapid adsorption kinetics and compatibility with vapor-phase processes.[18−26] However, SMIs often suffer from limited

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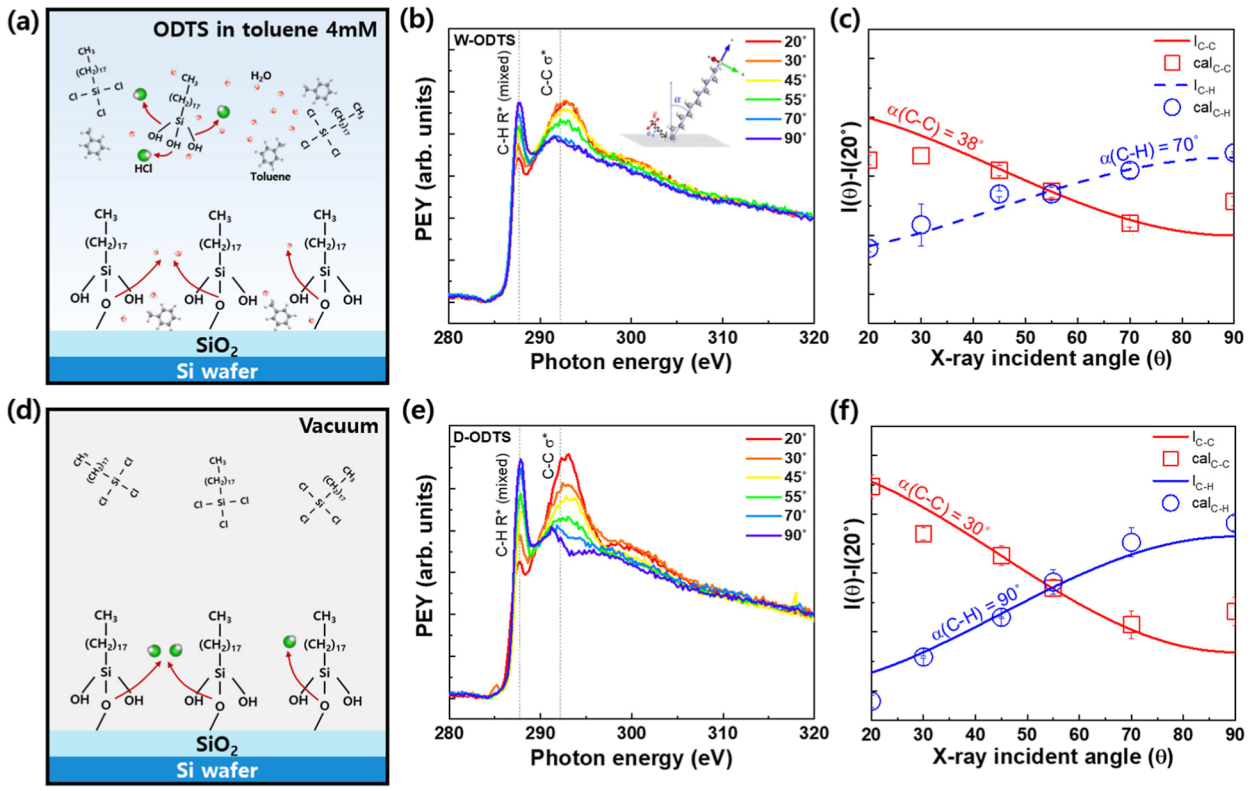


Figure 1. Deposition methods and NEXAFS characterization of ODTS SAMs on Si/SiO_2 substrates. (a) Schematic illustration of the solution-phased deposition procedures. (b) Angle-dependent C K-edge NEXAFS spectra obtained for solution-deposited ODTS. (inset) Diagram depicting the molecular tilt angle α and the photon incident angle θ. (c) Quantitative evaluation of the orbital angles of C−C σ* and C−H σ* bonds of solution-deposited ODTS. (d) Schematic illustration of the vapor-phased deposition procedures. (e) Angle-dependent C K-edge NEXAFS spectra obtained for vapor-deposited ODTS. (f) Quantitative evaluation of the orbital angles of C−C σ* and C−H R* (mixed) bonds of vapor-deposited ODTS.

surface stability and incomplete blocking due to their low molecular coverage and weak intermolecular interactions.

On OH-terminated oxide substrates, long-chain alkylsilanes such as octadecyltrichlorosilane (ODTS) form stable siloxane (Si−O−Si) bonds, effectively passivating the surface and suppressing precursor adsorption.[9−12,14,15] ODTS is therefore a representative inhibitor for AS-ALD on SiO_2 surfaces.[6,7,10,16,17]

Despite their widespread use, systematic correlation between molecular structure and inhibition performance in ODTS-based SAMs remains limited. In particular, the relationship between molecular tilt angle, grafted molecular areal density, and defect formation—which collectively govern precursor accessibility—has not been clearly established. While molecular orientation has been investigated using near-edge X-ray absorption fine structure (NEXAFS) spectroscopy,[14] its direct connection to surface coverage and ALD inhibition behavior has not been quantitatively resolved.

In practice, ODTS monolayers are predominantly formed via solution-phase deposition, which suffers from long processing times and increased susceptibility to defect formation. Vapor-phase deposition offers potential advantages in process integration and structural uniformity; however, its application to AS-ALD remains limited to a few studies.[12,27]

In this study, we investigate the structure−inhibition relationship of ODTS SAMs by combining angle-dependent NEXAFS spectroscopy with atomistic simulations. By linking experimentally measured tilt angles to grafted molecular areal density and monolayer structure, we establish a framework for interpreting molecular ordering in ODTS films. Here, grafted molecular areal density refers to the number of covalently anchored ODTS molecules per projected substrate area, rather than the fraction of all surface silanol groups chemically consumed. Through comparative analysis of solution- and vapor-deposited SAMs, we further demonstrate how molecular-level structural differences influence precursor accessibility and nucleation behavior during ALD. This framework extends the broader concept of structure-dependent SAM functionality to AS-ALD inhibition.

2. RESULTS/DISCUSSION

2.1. Structural Characterization of Octadecyltrichlorosilane Self-Assembled Monolayers: Wet vs Vapor Deposition

Figure 1 illustrates the structural characteristics of ODTS SAM in relation to the deposition method, and it demonstrates the ODTS molecular alignment analyzed through angle-dependent NEXAFS. Effective AS-ALD processes utilize surface passivation to suppress unwanted nucleation in nongrowth areas. Exploiting ODTS as a passivation layer, this study explored two deposition methods: solution-phase immersion (W-ODTS) and vapor-phase deposition (D-ODTS).

A Si wafer with a native oxide layer is submerged in a 4 mM ODTS solution prepared in toluene for 24 h at ambient temperature (refer to the Materials and Methods for further details). Through this immersion technique, hydrolyzed ODTS molecules can adsorb onto the hydroxylated surface and

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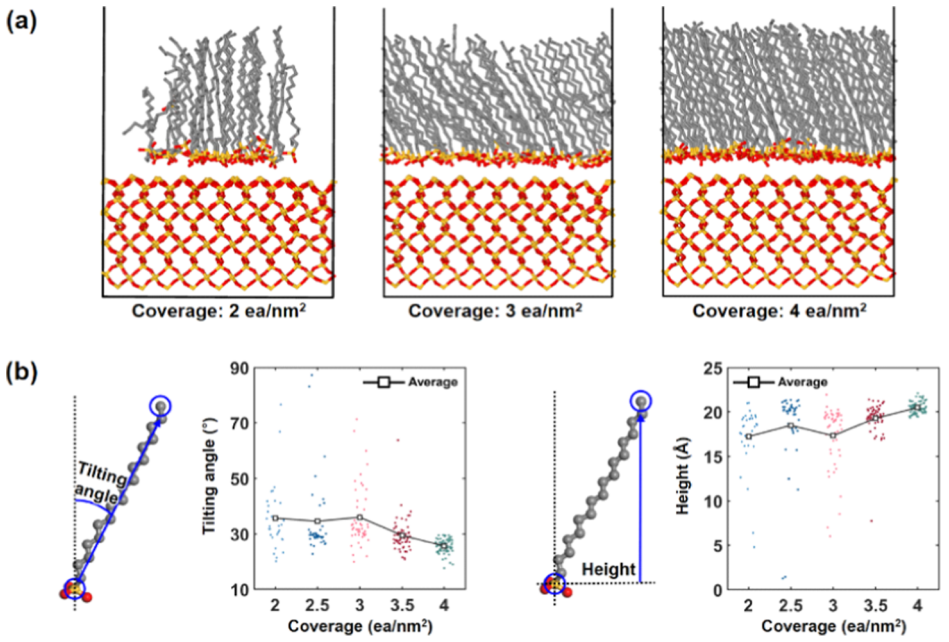


Figure 2. MD-derived relationship between molecular tilt angle, grafted molecular areal density and molecular height of ODTS monolayers on SiO_2. (a) All-atom models constructed at different surface coverages. (b) Variation of molecular tilt angle (left) and molecular height (right) as a function of surface coverage.

establish covalent bonding. However, residual solvent (toluene), water, and byproducts can lead to partial physisorption and molecular aggregation, reducing the alignment of the ODTS molecules (Figure 1a). Figure 1b displays angle-dependent NEXAFS spectra acquired in partial electron yield (PEY) mode, which provides molecular orientation information by measuring changes in absorption intensity as a function of the X-ray incidence angle. Although the effective probing depth in NEXAFS can vary with the X-ray incidence angle, this effect is negligible in the present study because the thickness of the ODTS monolayer (~1.7 nm) is significantly smaller than the probing depth in PEY mode (~2−3 nm). As a result, the entire monolayer is consistently probed across all incidence angles, and the observed angular dependence predominantly reflects molecular orientation rather than depth-dependent sampling effects. PEY-NEXAFS is becoming an increasingly vital tool for characterizing SAMs on surfaces in foundational AS-ALD research.

The average alkyl-chain axis of ODTS is treated as the *c*-axis, and the angle α is defined as the angle between this axis and the surface normal; the photon incidence angle is denoted by θ (Inset of Figure 1b). The resonance intensity reaches a maximum when the electric field polarization of the incident X-rays is aligned with the transition dipole moment (TDM) of the target orbital. The angular dependence of absorption intensity differs fundamentally depending on orbital symmetry, which falls into two cases: vector orbitals, whose TDM is oriented along a single axis, and plane orbitals, whose TDM is distributed within a plane. These two symmetry classes follow distinct angular expressions (see Supporting Information S1 for full derivations), and applying the wrong formula to a given resonance would systematically misestimate the molecular tilt angle.

In ODTS-based SAMs, prominent peaks at 287.7 and 292.9 eV correspond to transitions from the C 1s core level to the C−H Rydberg/σ^* mixed state (R*) and the C−C σ^* antibonding orbital, respectively, both of which show a clear angular dependence (Figures 1c,f and S2 and S3). Crucially, these two resonances belong to different symmetry classes. The C−C σ^* orbital at 292.9 eV is a vector orbital, with its TDM oriented along the molecular backbone; accordingly, the vector orbital expression was applied to extract the average tilt angle α. Because ODTS contains a long C18 alkyl chain, the extracted α should be interpreted as an effective, ensemble-averaged orientation of the alkyl backbone rather than the tilt of a perfectly rigid all-trans chain. Local conformational disorder, such as gauche defects or chain bending, may contribute to the angular distribution of C−C bonds but is not independently quantified in this analysis. The C−H R* band at 287.7 eV, by contrast, has plane orbital symmetry, with TDMs distributed within a plane perpendicular to the chain axis, and its angular dependence therefore follows the plane orbital formula (where $\beta = 90° - \alpha$). Quantitative analysis based on the C−C σ^* vector orbital resonance established an average tilt angle of $\alpha = 38°$ for W-ODTS. Independent analysis of the C−H R* plane orbital resonance yielded a consistent molecular orientation, confirming the validity of the approach.

The vapor deposition approach depicted in Figure 1d involves vaporizing ODTS and introducing the vapor into a vacuum chamber containing a Si wafer with a native oxide layer (refer to the Materials and Methods for details). This solvent-free technique reduces the risk of contamination and facilitates the formation of a cleaner ODTS layer with improved molecular alignment. D-ODTS monolayer formation is achieved within 3 min, and the extent of molecular orientation is influenced by the deposition duration (see Materials and Methods). For D-ODTS, the measured average tilt angle (α) was 30°, as shown in Figure 1f. To confirm the reproducibility of the tilt-angle analysis, angle-dependent C−C σ^* NEXAFS measurements were repeated for three independently prepared samples of each ODTS type; the normalized angular dependences and extracted tilt angles are summarized in Figure S4 and Table S1, respectively.

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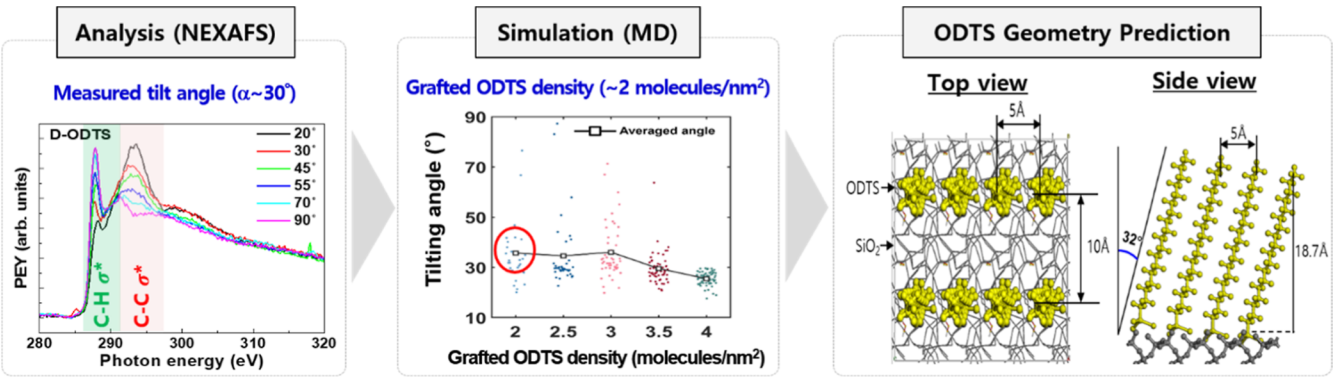


Figure 3. Proposed procedure for predicting the geometric structure of ODTS-based SAMs from the correlation between the NEXAFS-derived tilt angle and MD-derived ODTS areal density. The measured tilt angle $\alpha_{NEXAFS} \approx 30°$ corresponds to an ODTS areal density of ~2 molecules/nm^2, defined as the number of covalently anchored ODTS molecules per projected SiO_2 surface area, and yields a representative model geometry with $\alpha_{MD} \approx 32°$.

These observations show that vapor deposition yields improved molecular alignment compared to solution methods, highlighting its potential as an alternative route for AS-ALD applications. Additionally, we performed elemental analysis on D-ODTS and W-ODTS films using synchrotron-based X-ray photoemission spectroscopy (XPS) with $h\nu$ = 650 eV (Figure S5). Representative XPS analysis showed a lower O-related signal for D-ODTS than for W-ODTS, suggesting reduced contamination associated with solution processing under the present preparation conditions.

2.2. Modeling of Octadecyltrichlorosilane Surface Coverage

Atomistic simulation models of ODTS monolayers on a SiO_2 substrate were constructed using an all-atom force field to examine changes in tilt angle and molecular height as a function of surface coverage. In this work, the MD-derived coverage is defined as the grafted molecular areal density of ODTS, which corresponds to the number of covalently anchored ODTS molecules per projected SiO_2 surface area. This descriptor should not be interpreted as the fraction of all surface silanol groups that are chemically passivated, nor as the total area inaccessible to ALD precursors. Consequently, unreacted silanol groups remaining on the substrate are excluded from this definition of coverage. Figure 2a depicts the equilibrated configurations at different surface coverages, demonstrating that intermolecular van der Waals (vdW) interactions drive the formation of vertically aligned monolayer structures.

As shown in Figure 2b, MD simulations demonstrate a clear inverse relationship between molecular tilt angle and surface coverage, providing a practical basis for estimating surface coverage from experimentally measured tilt angles. This inverse relationship is well established for aliphatic SAMs, where increasing grafted molecular areal density strengthens lateral vdW interactions among alkyl chains and favors more upright average molecular configurations. In the present ODTS/SiO_2 system, the simulations consistently reproduce this trend and further reveal that orientational disorder becomes particularly pronounced at reduced coverage, especially near the edges of molecular clusters. This indicates that lateral intermolecular interactions play a dominant role in stabilizing vertical alignment, beyond the anchoring provided by covalent Si−O bonding to the substrate.

In contrast to the decreasing tilt angle, the effective molecular height increases with grafted molecular areal density, consistent with a more upright average chain configuration. The MD simulations further indicate that higher grafted molecular areal density reduces molecular-scale height variation and decreases nanoscale interstitial voids within the ODTS layer (Figures S6 and S7). These MD-derived descriptors suggest fewer precursor-accessible pathways through the ODTS layer, which is consistent with the improved ALD inhibition observed for D-ODTS. We note that the molecular-scale height variation obtained from MD is distinct from experimentally measured AFM RMS roughness, which was not obtained in this study.

2.3. Geometry Prediction of the Octadecyltrichlorosilane Monolayer

The geometric structure of ODTS monolayers can be inferred by combining NEXAFS-derived tilt angles with the MD-established relationship between tilt angle and grafted molecular areal density (Figure 3). Using this framework, the experimentally measured tilt angle of ~30° for vapor-deposited ODTS corresponds to an estimated effective grafted molecular areal density of ~2 molecules/nm^2 and an effective monolayer thickness of ~17 Å within the NEXAFS-MD correlation framework. In contrast, the larger tilt angle of ~38° observed for solution-deposited films implies a comparatively lower molecular density, reflecting a more disordered molecular arrangement.

These structural differences are further manifested in the effective molecular height: the lower tilt angle of D-ODTS corresponds to a more upright and extended average chain configuration and greater effective monolayer thickness, whereas the higher tilt angle of W-ODTS reflects increased chain disorder and a reduced effective thickness. Consequently, vapor-deposited ODTS monolayers are expected to exhibit fewer nanoscale interstitial voids and reduced molecular-scale height variation, both of which may influence precursor access during ALD.

This integrated structural analysis provides a direct basis for understanding the differences in ALD inhibition performance observed between the two deposition methods.

2.4. HfO_2 and Ru Area-Selective Atomic Layer Deposition on Octadecyltrichlorosilane Inhibitor Film

Following the formation of ODTS monolayers, the ALD inhibition performance was evaluated for both dielectric

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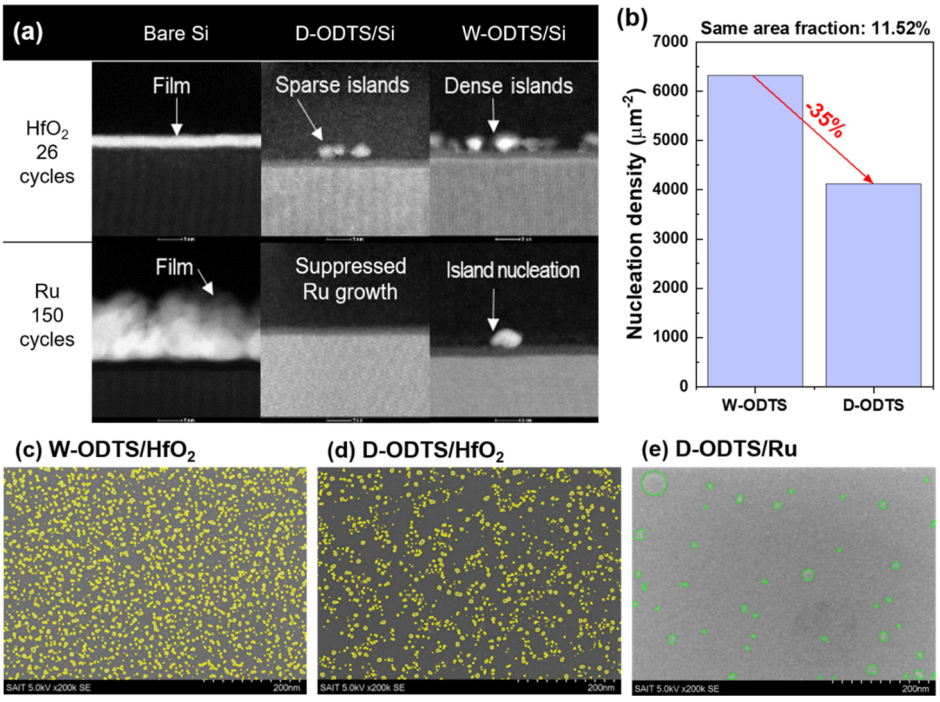


Figure 4. Nucleation behavior and inhibition performance of ODTS monolayers. (a) Cross-sectional SEM images of HfO_2 (top row) and Ru (bottom row) deposited on bare Si/SiO_2 (reference), W-ODTS, and D-ODTS after ALD. (b) Representative SEM image-based comparison of HfO_2 nucleation density on W-ODTS and D-ODTS surfaces at an identical projected area fraction of ~11.5%, showing a ~35% lower nucleation density on D-ODTS. (c,d) Top-view SEM images with segmented HfO_2 nuclei on W-ODTS and D-ODTS, respectively. Yellow features indicate the detected HfO_2 nuclei used for nucleation-density analysis. (e) Top-view SEM image with segmentation overlay of Ru deposited on D-ODTS. Green contours indicate rare isolated residual Ru nuclei detected within the field of view examined.

(HfO_2, 26 cycles) and metal (Ru, 150 cycles) deposition processes. Figure 4 summarizes the nucleation behavior on Si/SiO_2, W-ODTS/SiO_2, and D-ODTS/SiO_2 substrates.

For HfO_2 deposition, both W-ODTS and D-ODTS suppress nucleation compared to the Si/SiO_2 substrate; however, distinct differences arise depending on the SAM structure. As shown in the cross-sectional SEM images in Figure 4a, W-ODTS exhibits dense island formation, whereas D-ODTS results in sparse island nucleation, indicating more effective inhibition. To provide a representative image-based comparison of HfO_2 nucleation behavior, the nuclei density was evaluated at an identical projected area fraction of ~11.5%. Fixing the area fraction, rather than directly comparing samples after a given number of ALD cycles, minimizes the contribution of differences in lateral island growth and allows the comparison to focus on the number of nucleation events. Under this condition, D-ODTS exhibits a ~35% lower HfO_2 nucleation density than W-ODTS (Figure 4b), indicating a reduced density of nucleation-active sites on the D-ODTS surface. The detailed image-analysis parameters and quantitative values are summarized in Table S2. The corresponding top-view SEM images with segmented HfO_2 nuclei are shown in Figure 4c,d, and the raw SEM images are provided in Figures S8a and S8b.

This reduced HfO_2 nucleation density on D-ODTS is consistent with the structural characteristics identified by angle-dependent NEXAFS and MD simulations. As established in Section 2.2 and 2.3, D-ODTS exhibits a smaller molecular tilt angle and a higher effective areal density inferred from the NEXAFS-MD correlation. Although the MD-derived effective areal density does not directly quantify the fraction of chemically passivated silanol groups, the lower HfO_2 nuclei density on D-ODTS indicates fewer precursor-accessible nucleation-active sites during the initial ALD stage. The sparse HfO_2 nuclei remaining on D-ODTS suggest that a small population of residual accessible reactive sites is still present.

For Ru deposition, the inhibition behavior differs from that of HfO_2 and is discussed qualitatively rather than through a direct nucleation-density comparison, because Ru island formation on W-ODTS was too dense for reliable feature counting. In the cross-sectional SEM images, W-ODTS shows clear Ru island nucleation, whereas D-ODTS shows strongly suppressed Ru growth without a continuous Ru film or large Ru islands in the selected field of view (Figure 4a). To avoid overinterpretation of a local cross-sectional image, a top-view SEM and segmentation overlay of Ru deposited on D-ODTS is shown in Figure 4e, and the corresponding raw SEM image is provided in Figure S8c. The image-analysis parameters and auxiliary nuclei-density value for Ru/D-ODTS are also summarized in Table S2. The top-view result confirms strong suppression of Ru nucleation on D-ODTS, with only rare isolated nuclei detected within the examined field of view.

The different nucleation behaviors of HfO_2 and Ru on D-ODTS can be attributed to their process-specific ALD chemistry. In the HfO_2 process, nucleation can still occur at residual precursor-accessible defects, such as nanoscale voids or locally exposed hydroxylated sites, even on the relatively compact D-ODTS monolayer. Therefore, sparse HfO_2 nuclei are still observed. In contrast, Ru nucleation is more strongly suppressed on the methyl-terminated D-ODTS surface, likely because the alkyl-chain layer limits precursor adsorption and defect-mediated reaction pathways. Thus, sparse HfO_2 nucleation and strongly suppressed Ru growth on D-ODTS are not contradictory, but reflect the different sensitivity of the

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oxide and metal ALD processes to residual nucleation-active sites.

Overall, these results demonstrate that the improved inhibition performance of vapor-deposited ODTS is correlated with its more upright average molecular orientation and higher effective areal density. These structural characteristics are associated with fewer precursor-accessible pathways, leading to improved ALD blocking performance for both oxide and metal deposition systems.

3. CONCLUSIONS

In this study, we demonstrate a rapid vapor-phase deposition route for forming ODTS blocking layers with improved molecular alignment on SiO_2 within minutes. Angle-dependent NEXAFS analysis reveals a reduced molecular tilt angle for vapor-deposited ODTS (~30° vs ~38° for solution-deposited films), indicating a more upright average molecular orientation than solution-deposited ODTS.

By combining NEXAFS measurements with atomistic simulations, we establish a clear relationship between molecular tilt, surface coverage, and monolayer structure, enabling semiquantitative interpretation of average molecular orientation and effective grafted molecular areal density from spectroscopic measurements. This structural framework provides insight into how molecular-level ordering governs nucleation behavior during ALD.

Correlating these structural characteristics with ALD experiments, we show that vapor-deposited ODTS significantly suppresses nucleation for both HfO_2 (~35% reduction in nucleation density) and Ru (strong suppression of nucleation), attributable to reduced nanoscale voids and defect-mediated adsorption sites. These findings identify molecular tilt angle and effective molecular areal density as useful structural descriptors that control inhibition efficiency in SAM-assisted AS-ALD.

Overall, this work establishes a structure−inhibition relationship in ODTS-based systems and identifies molecular tilt angle as a practical structural descriptor for rational SAM inhibitor design, highlighting vapor-phase SAM formation as a scalable strategy for high-performance area-selective deposition in advanced semiconductor fabrication.

4. MATERIALS AND METHODS

4.1. Preparation of W-Octadecyltrichlorosilane and D-Octadecyltrichlorosilane

Si wafers (200 mm diameter) with a native oxide layer of approximately 1−5 nm thickness were employed as substrates. Octadecyltrichlorosilane (ODTS, >97%) was sourced from Sigma-Aldrich (USA), and RCA cleaning was carried out to eliminate organic residues from the silicon wafer surfaces. For both W-ODTS and D-ODTS samples, fabrication protocols were as follows. For W-ODTS: the Si wafers were immersed in a 4 mM ODTS solution in pure toluene at ambient temperature for 24 h to promote SAM assembly. For D-ODTS: vapor-phase deposition of ODTS was undertaken using a commercial thermal evaporator (HVC-2050DA, Hanil Vacuum Co., Ltd., South Korea). Approximately 0.5 g of ODTS was placed in a stainless-steel vessel packed with steel wool (tablet). The Si wafers were positioned on a mounting jig, while the tablet was set on a tungsten boat within the vacuum chamber. When the chamber pressure reached 2.0×10^{-5} Torr, the tungsten boat was electrically heated to vaporize the ODTS molecules. The resulting vapor subsequently physisorbed onto the wafer surfaces, with deposition durations in the range of 100−300 s. A quartz crystal microbalance (QCM) integrated into the evaporator chamber was utilized to monitor both the deposition rate and film thickness in real time. Following SAM formation, both W-ODTS and D-ODTS wafers were sonicated in toluene for 10 min to remove any physisorbed material. Subsequent steps involved rinsing the wafers with isopropyl alcohol (IPA) and drying them under a nitrogen gas stream.

4.2. Atomic Layer Deposition Process

ALD blocking tests were conducted on two ODTS-treated samples and a reference bare silicon wafer using the Lucida C200-PL system (NCD Co., Ltd., South Korea). The HfO_2 ALD process operated at 150 °C for 26 cycles, employing tetrakis(ethylmethylamino)hafnium (TEMAHf) as the precursor and water as the counter-reactant. Ru ALD was carried out at 200 °C for 150 cycles, using ethylbenzene ethyl-1,4-cyclohexadiene ruthenium (EBECHRu) as the precursor, with O_2 serving as the counter-reactant. In both ALD processes, precursor and counter-reactant pulse times were set at 1.5 and 0.5 s, respectively, followed by 20 s of Ar purging. Growth rates of HfO_2 and Ru on Si wafers were determined to be 0.96 and 0.51 Å/cycle, respectively.

4.3. Surface Characterization and Near-Edge X-Ray Absorption Fine Structure Studies

To confirm SAM deposition and assess film wettability, static water contact angle (WCA) measurements were performed on the ODTS-treated samples using a DSA100 contact angle analyzer (KRUSS, Germany). Cross-sectional TEM specimens were prepared with a dual-beam FIB-SEM system (Helios NanoLab 450HP, FEI, USA) and examined at 200 kV using a Titan G2 system (Thermo Fisher Scientific, USA).

XPS and angle-dependent NEXAFS spectroscopy were performed at the 4D PES beamline of the Pohang Accelerator Laboratory (PAL, South Korea) to investigate the surface arrangement and chemical states of the SAM molecules. The beamline operates with a degree of linear polarization of $P = 0.8$. The samples were initially placed into a vacuum chamber (base pressure = 4.0×10^{-8} Torr) connected to an analysis chamber (base pressure = 3.0×10^{-10} Torr) that housed both XPS and NEXAFS detectors at the same focal point.

For survey scans in XPS and NEXAFS, the photon incident angle was set to 45°, and photon energy was calibrated using Au 4*f* as a reference. NEXAFS spectra were recorded in both total electron yield (TEY) mode, with the sample grounded and no retarding voltage applied, and PEY mode with a probing depth of approximately 3 nm, the latter of which was used for the primary SAM analysis due to its enhanced surface sensitivity. Angular dependence was assessed by varying the angle between the incident X-ray beam and the sample surface (20°, 30°, 45°, 55°, 70°, and 90°). The synchrotron radiation was linearly polarized, with the electric field vector (E-vector) lying predominantly in the orbital plane of the storage ring. As a result, the NEXAFS intensity is governed by the dot product between the E-vector and the TDM of the probed molecular orbital: intensity is maximized when they are parallel and vanishes when perpendicular. For the ODTS alkyl chain, the relevant transitions are σ^* resonances (C−H and C−C), whose TDMs are oriented along the respective bond axes. The systematic variation of NEXAFS intensity with incident angle therefore directly reflects the average orientation of these bonds relative to the surface normal, enabling quantitative determination of the molecular tilt angle.

4.4. Time Dependence of D-Octadecyltrichlorosilane

To understand and optimize the MLD process using the novel tablet-impregnated ODTS inhibitor evaporation method, a conventional QCM instrument was utilized. The QCM crystal was coated with gold to ensure accurate monitoring. During deposition, vaporized ODTS molecules reacted with hydroxyl groups on the native oxide-covered Si substrate, forming a Si−O−Si chemical bond and creating a monolayer. As deposition time increased, a progressively thicker physisorbed overcoating formed on the monolayer. Detailed investigation of ODTS film formation as a function of process time using QCM revealed that the film thickness initially increased, peaking at approximately 490 Å before gradually diminishing to tens of Å after prolonged deposition times. Simultaneously, the WCA

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reduced from approximately 110° at 100 s of deposition to about 85° at 300 s (Figure S9). The high initial WCA of about 110° indicated successful surface functionalization with hydrophobic ODTS molecules. Under vacuum conditions, weakly physisorbed ODTS molecules tend to desorb from the substrate. For comparative purposes, a solution-deposited ODTS sample, prepared using a conventional immersion technique, achieved a WCA of approximately 108°.

4.5. MD Simulations

All-atom modeling of SiO_2/ODTS unit cells was performed using BIOVIA Materials Studio 2020 (Dassault Systémes, France), while model relaxation and structural characterization were carried out using the open-source MD code LAMMPS (Sandia National Laboratories, USA).[28] All interatomic interactions were described using the polymer-consistent force field (PCFF) developed by Sun.[29] A supercell with dimensions of 3.8 × 3.8 × 1.8 nm^3 was constructed by replicating the atomic lattice of alpha quartz (SiO_2). Periodic boundary conditions were applied along the *x*- and *y*-axes, and a sufficiently thick vacuum layer was added along the *z*-axis to expose the free surface. ODTS, deployed as the coating material for forming SAM, exhibits a linear structure with 18 hydrocarbon backbone units. Models containing 28−56 ODTS molecules on the SiO_2 surface were analyzed under surface coverage conditions of 2−4 molecules/nm^2. Initially, the ODTS molecules were held together by vdW forces before forming covalent bonds with the SiO_2 surface. The assembled SiO_2/ODTS systems were equilibrated under the *NVT* ensemble at 300 K for 0.5 ns (Figure S10), utilizing a 0.1 fs time step and the Verlet algorithm for numerical integration.

After achieving equilibrium, analysis of the tilt angle distribution of ODTS molecules was conducted. The tilt angle was determined by calculating a vector from the silicon atom of the silane headgroup to the terminal carbon atom, and then measuring the angle between this vector and the normal to the SiO_2 surface (*z*-axis). The effective monolayer height was evaluated by taking the *z*-axis distance between the terminal carbon and silicon atoms.

4.6. Quantitative Analysis of Nucleation Density from SEM Images

Nucleation density, projected area fraction, and equivalent diameter were quantified from top-view SEM images of Ru and HfO_2 using custom Python scripts based on the OpenCV and NumPy libraries. Image preprocessing included noise reduction and contrast enhancement to improve feature visibility prior to segmentation.

For HfO_2 samples, a simplified segmentation approach was employed due to the more continuous and film-like growth morphology. Images were first smoothed using a Gaussian filter (kernel size: 5 × 5 pixels). To capture both small nuclei and larger film-like regions, a dual-threshold binarization scheme was applied: a lower threshold (intensity = 28) was used to detect fine features, while a higher threshold (intensity = 150) was used to capture larger, high-contrast regions. The resulting binary masks were combined using a logical OR operation to generate the final segmentation mask.

For Ru samples, grayscale images were first denoised using a median filter (kernel size: 17 × 17 pixels) to suppress high-frequency noise while preserving edge features. A morphological top-hat transformation with a 51 × 51 pixel elliptical structuring element was subsequently applied to enhance bright nuclei features against the nonuniform background. Local contrast was further improved using contrast-limited adaptive histogram equalization (CLAHE; clip limit = 4.0, tile grid size = 8 × 8). The contrast-enhanced image was binarized using a fixed intensity threshold of 40, which was empirically optimized and consistently applied across all data sets. Binary mask refinement was performed via sequential morphological opening (1 iteration) and closing (2 iterations), both with a 7 × 7 elliptical structuring element, to remove isolated noise pixels and fill small gaps within nuclei boundaries.

Individual features were identified via external contour extraction from the refined binary masks. The uppermost 30-pixel region containing the SEM scale bar annotation was excluded from the analysis. Detected objects were filtered by projected area (130−20,000 $pixels^2$) for both materials. For Ru samples, an additional circularity criterion ($C = 4\pi A/P^2$), where A is the projected area and P is the perimeter) was applied to exclude elongated artifacts. A two-tier circularity threshold was implemented: objects with area <250 $pixels^2$ required C > 0.4, whereas larger objects required C > 0.1 to accommodate partially coalesced nuclei. This additional filtering reflects the discrete island-like nucleation behavior of Ru. In contrast, circularity filtering was not applied to HfO_2 samples due to their more continuous growth morphology.

After exclusion of the scale-bar annotation region, the valid analyzed image size was 1280 × 895 pixels for each SEM image. A pixel-to-physical scale conversion was established from the SEM scale bar (200 nm per 403 pixels; 0.4963 nm $pixel^{-1}$). Nucleation density was calculated as the number of detected features divided by the valid analysis area, expressed in units of μm^{-2}. The projected area fraction was determined as the ratio of feature-occupied pixels to the total valid analysis area. The equivalent diameter was computed as $d_{eq} = \sqrt{4A/\pi}$, where A is the projected area of each segmented feature, to approximate the effective lateral size of individual features. The equivalent diameter was then converted to nanometers using the pixel size calibration. All analyses were performed using Python 3.

■ ASSOCIATED CONTENT

Data Availability Statement

All the Supporting Information are provided in the main text and Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.langmuir.6c02347.

> Theoretical framework for angle-dependent NEXAFS analysis, reproducibility of NEXAFS tilt-angle fitting, additional C K-edge NEXAFS spectra and intensity contour plots, XPS profiles of W-ODTS and D-ODTS, nanosized-void distribution and ODTS monolayer surface visualization from MD simulations, top-view SEM images and segmentation analysis of ALD nucleation, SEM image-based quantification parameters, in situ QCM thickness and WCA measurements, all-atom SiO_2/ODTS MD simulation model details (PDF)

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Author Contributions

B.P. initially designed the project and wrote the manuscript. H.K. and J.C. calculated the MD simulations and authored this section. E.S., J.L., S.K., and K.I. conducted the spectroscopy analysis and composed this section. E.S. substantially contributed to the overall manuscript revision, including restructuring the manuscript and refining the scientific interpretation. S.H. and M.K. performed all sample preparation and characterization. All authors contributed to the overall scientific interpretation and revision of this manuscript.

Notes

The authors declare no competing financial interest.

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